

ADITYA BAYU PRATAMA

Supervisor MES & IT Engineer

SMART MANUFACTURING · INDUSTRY 4.0 · ENTERPRISE SYSTEMS

7+

YEARS IN SMART
MANUFACTURING

500+

PRODUCT MODELS
GOVERNED BY MES

5

ENGINEERS LED
& MENTORED

2×

GLOBAL 1ST PLACE
VE AWARD

CONTACT

LOCATION

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PORTFOLIO

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CORE COMPETENCIES

SOFTWARE

C# (.NET), SQL / MySQL, Python, React.js, JavaScript, C++, LabVIEW, Dart / Flutter, REST APIs

SMART MANUFACTURING

MES architecture, Industrial IoT, traceability, Pokayoke, AOI machine vision, SMT process, Node-RED

HARDWARE

PLC (BNSP certified), SCADA, Raspberry Pi, Arduino, Lidar, data acquisition, PID control

AI & TOOLING

AI-assisted development, prompt engineering, LLM integration, LM Studio, Claude Code, Cursor, Codex, OpenCV

QUALITY

FMEA, DMAIC, Six Sigma practices, 5M change management, value engineering, BOM & QCPC, NPI

LEADERSHIP

Team recruitment, department budget, network infrastructure, server maintenance, incident response

PROFILE

Supervisor MES & IT Engineer with more than seven years designing and running Manufacturing Execution Systems, Industrial IoT and machine vision for high-precision electronics and oil & gas. Grew the software and IT function from a solo role into a five-engineer department, architected a custom C#/.NET MES governing 500+ product models, and turned manual factory workflows into digital ones — one of them from 5–7 hours with 4–6 people to a 30-minute automated flow.

EXPERIENCE

GIKEN PRECISION INDONESIA

Batam · Apr 2019 – Present

Supervisor MES & IT Engineer

Jul 2026 – Present

- Lead the MES and IT engineering function: system roadmap, architecture standards, project prioritisation across production and support, department budget, and development of the engineering team.
- Own the Giken One platform strategy — MES, traceability, feeder asset control and the internal web applications the plant depends on.
- Report department performance to management: uptime, ticket resolution, project progress and savings from digitalisation.

Lead Software & IT Engineer

Apr 2019 – Jun 2026

- Designed and deployed a custom Windows-based Manufacturing Execution System (C#/.NET, MySQL) governing 500+ product models across automotive, medical and consumer electronics lines, enforcing Pokayoke defect prevention and CTQ quality compliance.
- Built Giken One, an enterprise web platform (React.js, MySQL) digitising employee records, org charts, overtime approvals, IT ticketing and project scheduling.
- Engineered an automated ESD wristband monitoring gate (Arduino, C#, MySQL) enforcing 100% compliance before line entry and tracking wristband defect rates.
- Automated QR-code thermal label printing across new electronic lines through a direct C# and SQL printer interface, eliminating printing delays.
- Established centralised network file servers across all departments, removing the risk of data loss from single-drive failure.
- Recruited, mentored and scaled the software and IT engineering department from a solo role into a five-engineer team.
- Contained a worm malware outbreak across 20 laptops within 3 hours with no production downtime; sustained 99.9% uptime on core MySQL servers; resolved factory network outages inside 15-minute SLAs.

EDUCATION

State Polytechnic of Batam

JUL 2017 – JUL 2021

Bachelor of Applied Science (S.Tr.T.)
Mechatronics, Robotics & Automation
Engineering

Graduated Cum Laude · GPA 3.81 / 4.00

CERTIFICATIONS

Google Data Analytics Professional

COURSERA · 2022

PLC Competency Certificate

BNSP INDONESIA · 2019

Electrical Power Installation Technician

LPJK · 2020

DevOps, Networks, Dart & Flutter

DICODING · 2023

HONORS

2025

1st Place — Giken Global Value Engineering

Intelligent Machine Data: real-time
analytics and line efficiency.

2023

1st Place — Giken Global Value Engineering

Manufacturing Traceability System:
eliminated defect passage between
processes.

EXPERIENCE (CONTINUED)

GIKEN PRECISION INDONESIA

Batam · Apr 2019 – Sep 2021

Software Test Engineer

Apr 2019 – Apr 2024

- Developed machine testing software for new electronic products and ran production trials before mass production.
- Translated customer requirements into test criteria and validated measurement accuracy between test equipment, PLCs and the MES database.

New Product Introduction Engineer

Apr 2019 – Sep 2021

- Studied new electronics projects from customer drawings to line readiness.
- Created and analysed FMEA, 5M change management, BOM lists and QCPC documents; ran trials and solved production problems with quality and process engineering.

CLADTEK BI-METAL MANUFACTURING

Batam · Apr 2018 – Jan 2019

Junior Programmer — R&D Department

Apr 2018 – Jan 2019

- Built an autonomous pipe-inspection robot (C#, Raspberry Pi, Lidar, light sensors) navigating inside industrial pipes, streaming live video and calculating weld defect location and severity.
- Developed low-latency transmission protocols connecting embedded sensors to central servers for real-time welding QA.

SELECTED PROJECTS

Manufacturing Traceability System — Giken One MES

2019 – Present

Plant-wide MES with per-line production dashboards, full unit traceback and SMT refill control across 500+ product models. 1st place globally, Giken Value Engineering Award 2023.

Intelligent Machine Data

2025

Real-time machine data acquisition and analytics replacing manual reporting with live line-efficiency and downtime visibility. 1st place globally, Giken VE Award 2025.

Auto Feeder Machine Data Generator

2026

Digitalised the SMT feeder data flow across four roles: 5–7 hours with 4–6 people became a 30-minute automated flow, cutting cycle time by roughly 93%.

PCBA Management Apps

2026

React.js and MySQL platform running the PCBA IT department: tickets, tasks, schedules, projects, attendance, assets and knowledge base on one live dashboard.

SMT Pick-Up Material Analytics Dashboard

2026

Automated analysis of pick-up data across 23 SMT lines: 7.9 million pickups, error rate and spoilage PPM per line and machine, plus high-risk feeder detection.

Digital Feeder Inventory & Tracking

2026

Real-time feeder location and asset control linking the pick-up material database to the refill system, down to area, line, machine, table and feeder number.

LabVIEW AOI PCB Inspection System

2021

Automated Optical Inspection in LabVIEW Vision detecting missing or misaligned components on PCB assembly lines, reducing manual QA headcount.